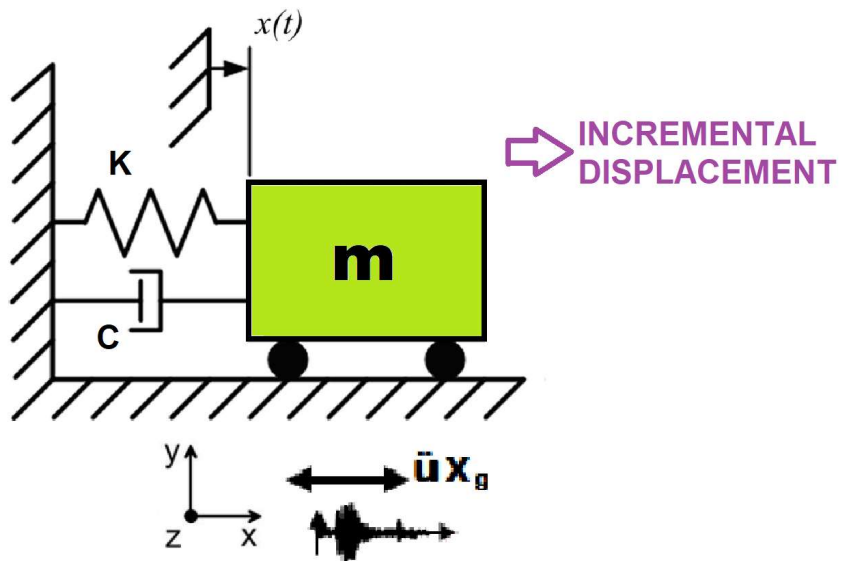


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

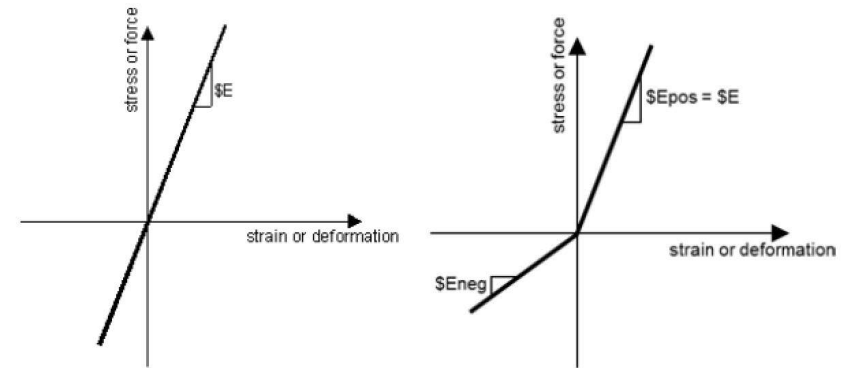
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

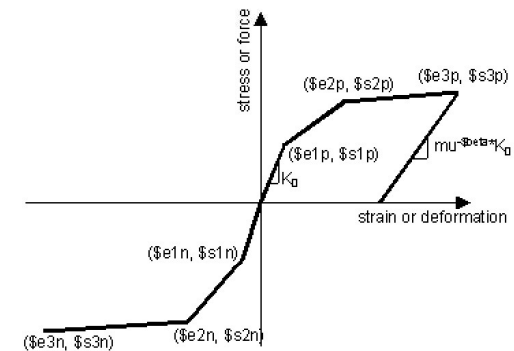
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

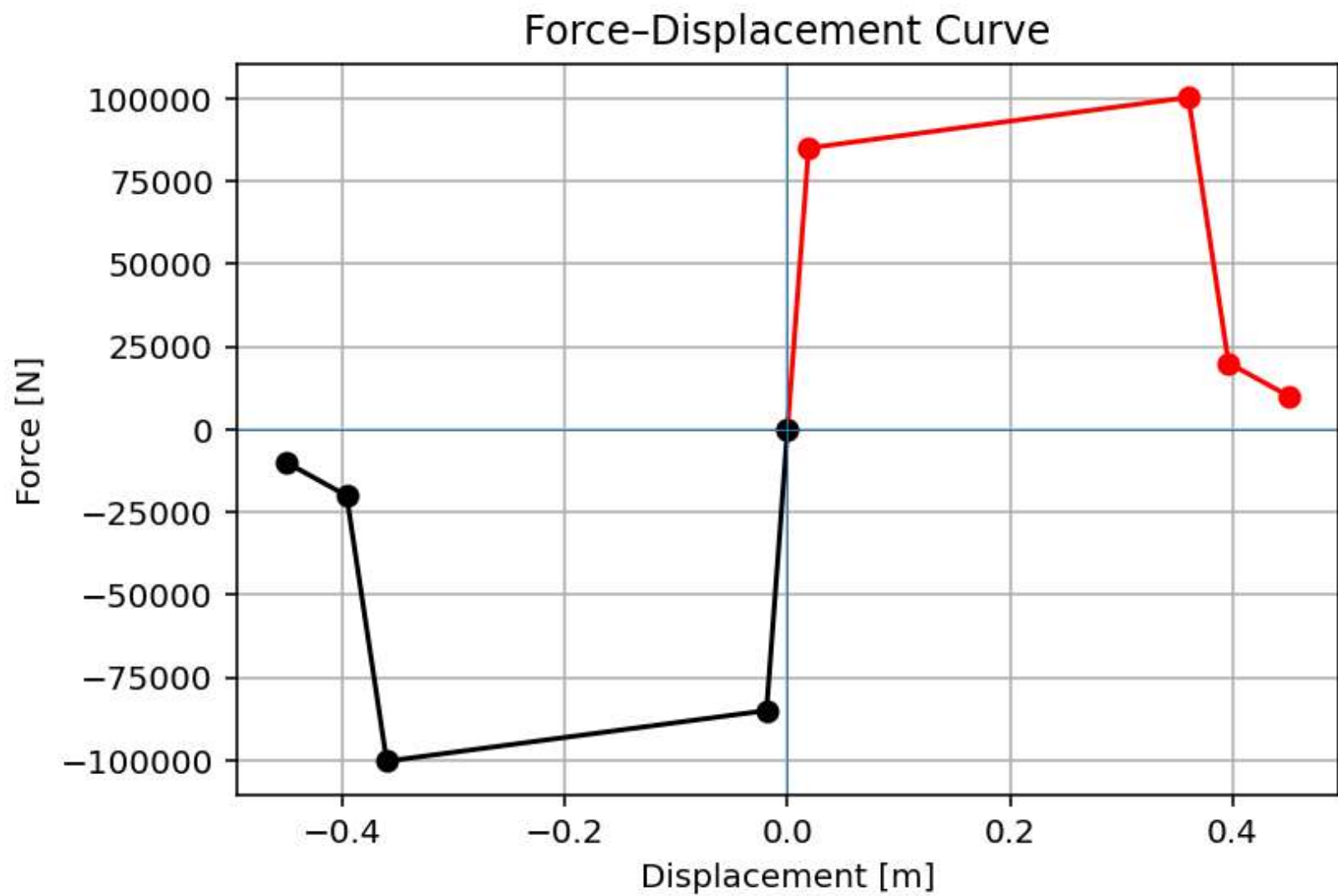
$$m\ddot{u} + c\dot{u} + ku = P(t)$$

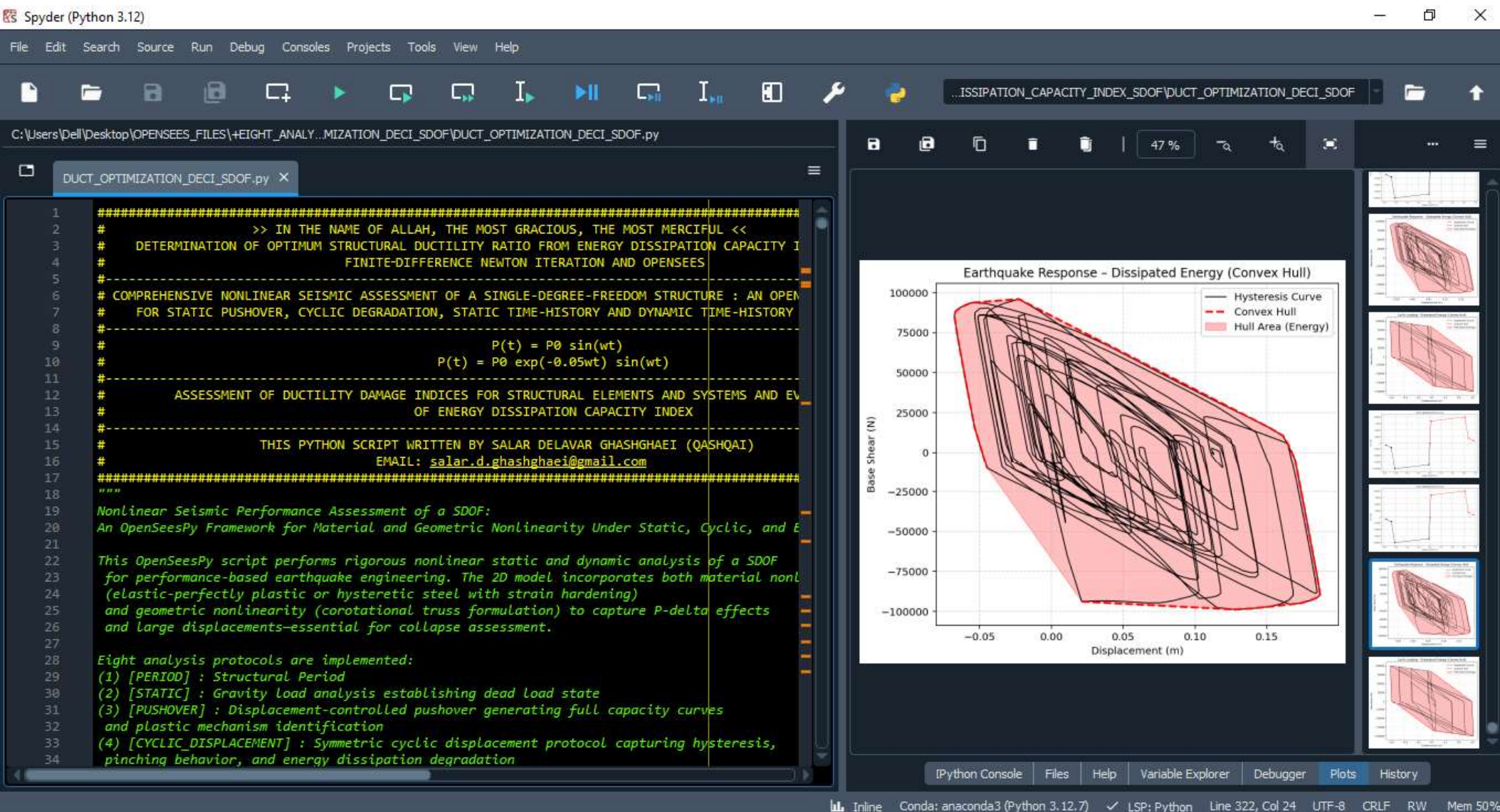


ELASTIC MATERIAL



INELASTIC MATERIAL





OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MIZATION_DECI_SDOF\DUCT_OPTIMIZATION_DECI_SDOF.py

DUCT_OPTIMIZATION_DECI_SDOF.py DISSIPATED_ENERGY_WITH_PLOT_FUN.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF OPTIMUM STRUCTURAL DUCTILITY RATIO FROM ENERGY DISSIPATION CAPACITY I
4 # FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
5 #
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPEN
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
8 #
9 # P(t) = P0 sin(wt)
10 # P(t) = P0 exp(-0.05wt) sin(wt)
11 #
12 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EV
13 # OF ENERGY DISSIPATION CAPACITY INDEX
14 #
15 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
16 # EMAIL: salar.d.ghashghaei@gmail.com
17 #####
18 """
19 Nonlinear Seismic Performance Assessment of a SDOF:
20 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and E
21
22 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
23 for performance-based earthquake engineering. The 2D model incorporates both material nonl
24 (elastic-perfectly plastic or hysteretic steel with strain hardening)
25 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
26 and large displacements-essential for collapse assessment.
27
28 Eight analysis protocols are implemented:
29 (1) [PERIOD] : Structural Period
30 (2) [STATIC] : Gravity load analysis establishing dead load state
31 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
32 and plastic mechanism identification
33 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
34 pinching behavior, and energy dissipation degradation
```

Console 1/A

```
+-----+
| lambda | omega | period | frequency |
+-----+
| 9.000e+02 | 30.0000 | 0.2094 | 4.7746 |
+-----+

Time: 20.0000, Displacement: 0.1123 mm, Reaction: -20.00 N

SEISMIC ANALYSIS DONE.

Exact Damping Ratio: 5.97731447e-03
Xmax: 32.53044849579263
Dissipated Energy from Earthquake= 31986.11 N·m
Dissipated Energy from Cyclic Displacement= 159935.60 N·m

DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]

ZONE 2: MINOR DAMAGE

Fmax: -0.0006336555210779693
DF: -0.6336833235387473
DX: -1.2654525055268235e-08
IT: 9 - RESIDUAL: 1.2654525055268235e-08 X: 32.529448495792636

Optimum X: 32.5294
Iteration Counts: 9
Convergence Residual: 1.2654525055e-08

Total time (s): 83.6250
```

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 146, Col 24 UTF-8 CRLF RW Mem 50%

Dissipated Energy from Earthquake= 31986.11 N·m

Dissipated Energy from Cyclic Displacement= 159935.60 N·m

DISSIPATED ENERGY CAPACITY INDEX: 19.999 [%]

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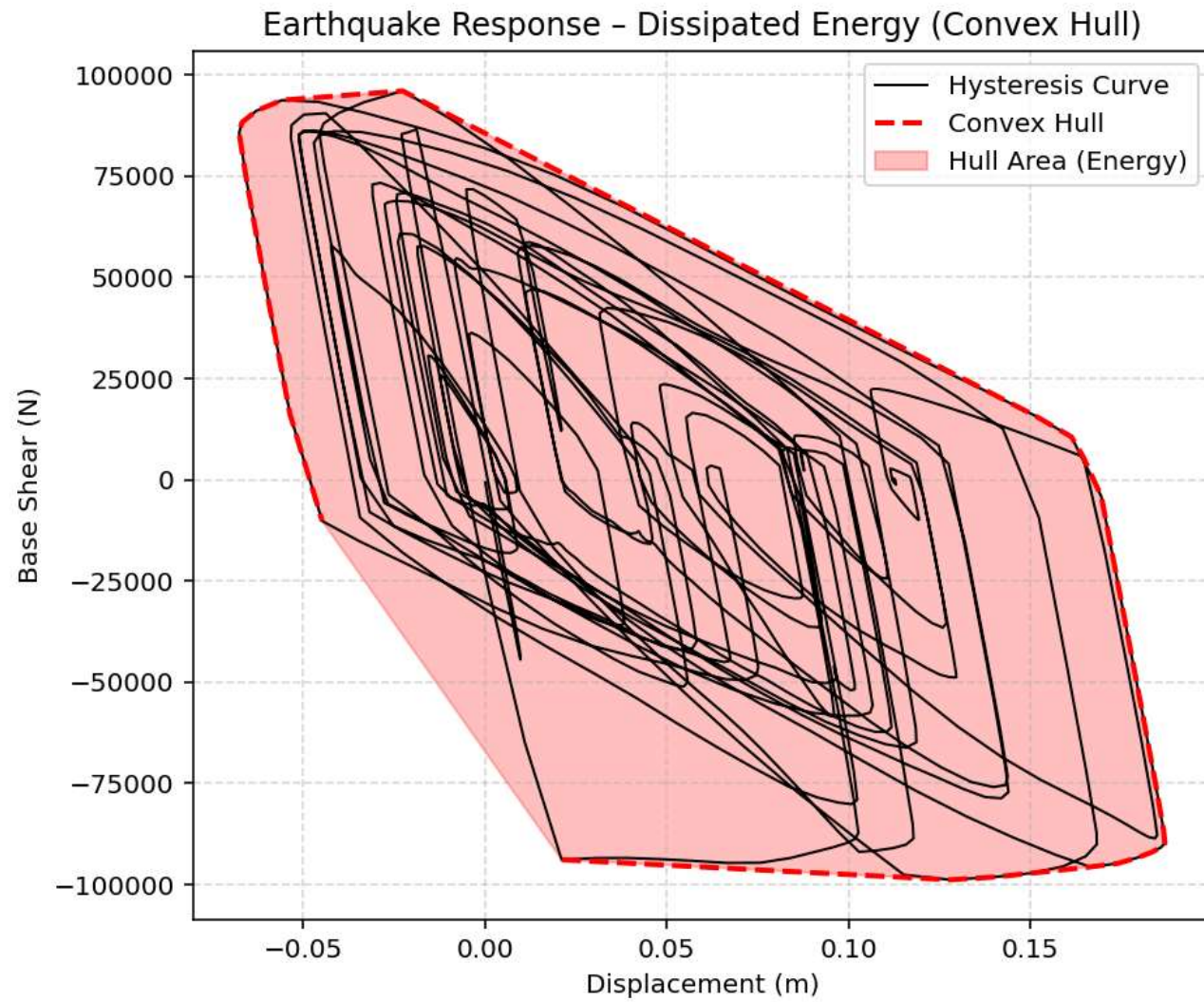
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Cyclic Loading - Dissipated Energy (Convex Hull)

